HESTIA

MICRO-PROJECT REPORT

submitted by

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to

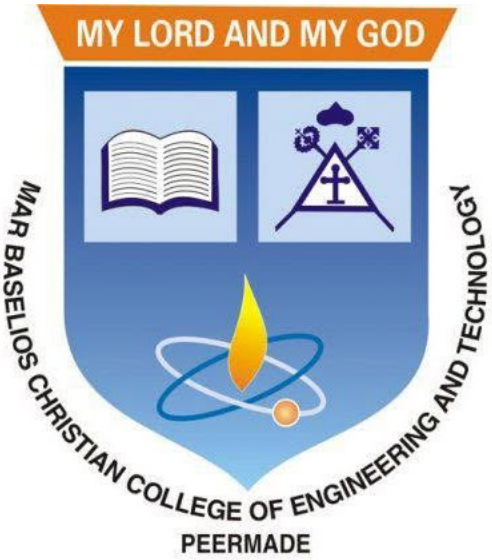
APJ Abdul Kalam Kerala Technological University

in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

in

Computer Science and Engineering (AI)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE

OCTOBER, 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE

CERTIFICATE

This is to certify that the micro project report entitled "**HESTIA**" submitted by **Nripan Satheesh (MBC25CA063)** and **Muhammed Afrithkhan (MBC25CA057)** to **APJ Abdul Kalam Kerala Technological University** in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Technology in Computer Science and Engineering (AI)** is a bonafide record of the micro project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Project Guide:

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- *Job Radar Service:* Filters corporate vacancies by district, minimum CGPA, and salary packages (LPA).
- *Frontier AI Inference Core:* Manages Groq Cloud Llama-3.3-70B prompt orchestration and multiturn conversational memory logging.

4. **Persistence Tier:** A relational SQL database operating in Third Normal Form (3NF) managed via Spring Data JPA with connection pooling.

3.4 SYSTEM ARCHITECTURE DIAGRAM

The high-level enterprise architecture diagram illustrating the interaction between the Presentation Layer, API Gateway, Verification Facade, Frontier AI Core, and Relational Persistence Tier is presented below:



Fig 3.1: Enterprise Architecture of Hestia Platform

As depicted in Fig 3.1, client requests transit the secure API Gateway before dispatching to dedicated service facades, ensuring robust data isolation and zero single points of failure.

3.5 CONCEPTUAL RELATIONAL DATABASE DESIGN

The persistence model of Hestia is designed in Third Normal Form (3NF) to eliminate data redundancy and preserve referential integrity across student records, academic grade cards, verification logs, and AI conversation histories.

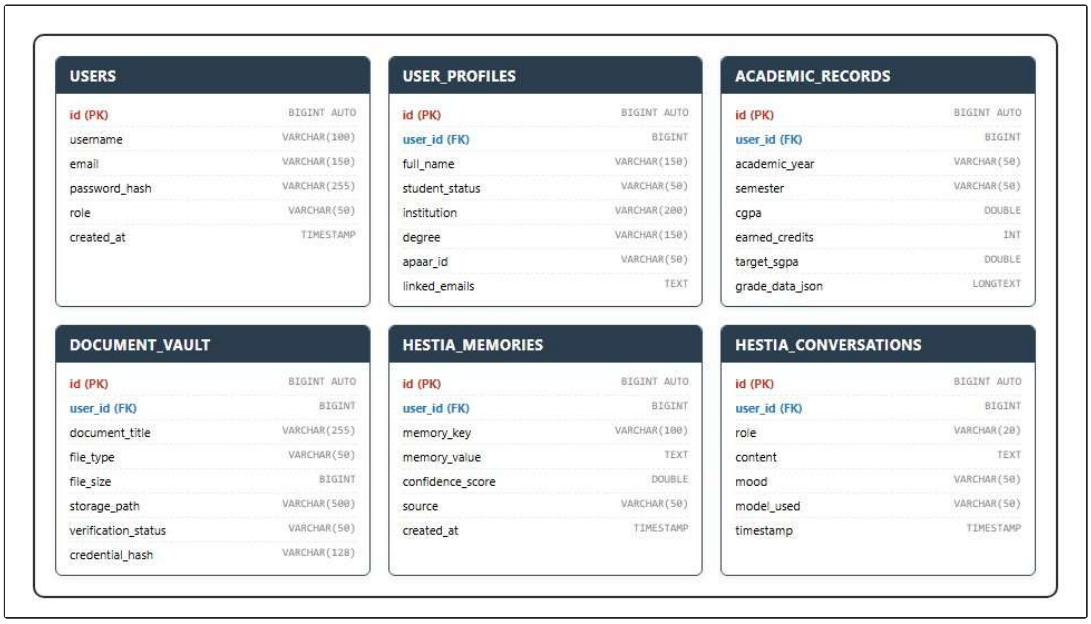


Fig 3.2: Relational Database Schema & Entity Relationships

Primary relational entities include Users, AcademicRecords, VerificationAudits, JobPostings, and AIMemoryVault, maintaining strict foreign key cascade rules and optimized B-Tree indexation.

3.6 STRENGTH OF THE PROPOSED SYSTEM

The architecture of Hestia offers distinct technical strengths over conventional university software:

- **Cryptographic Forensic Authenticity:** Prevents credential counterfeiting through automated threetier validation with zero reliance on manual paperwork.
- **Predictive CGPA Engineering:** Empowers students to model precise semester GPA requirements to meet corporate cut-offs.
- **Stateful AI Intelligence:** Delivers continuous, context-aware interview coaching through persistent neural memory logging.
- **High Throughput & Cloud Scalability:** Stateless Spring Boot backend enables horizontal autoscaling and sub-200ms API response latency.

SCREENSHOTS

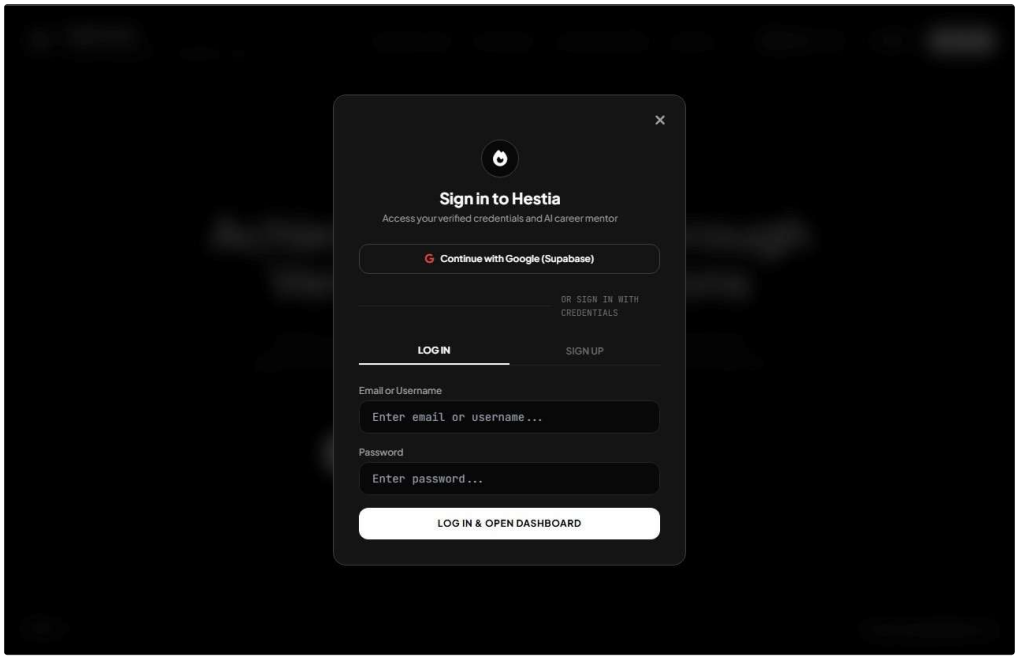


Fig 6.1: Hybrid Authentication Modal (Google SSO & Password Sync)

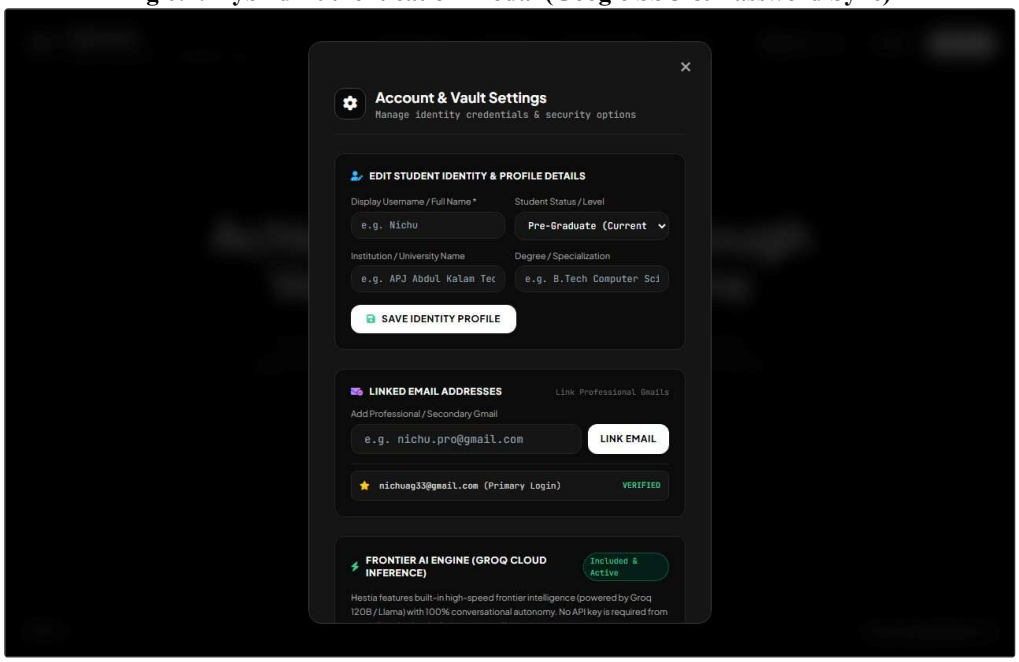


Fig 6.2: Student Identity Profile & Multi-Linked Email Settings Modal

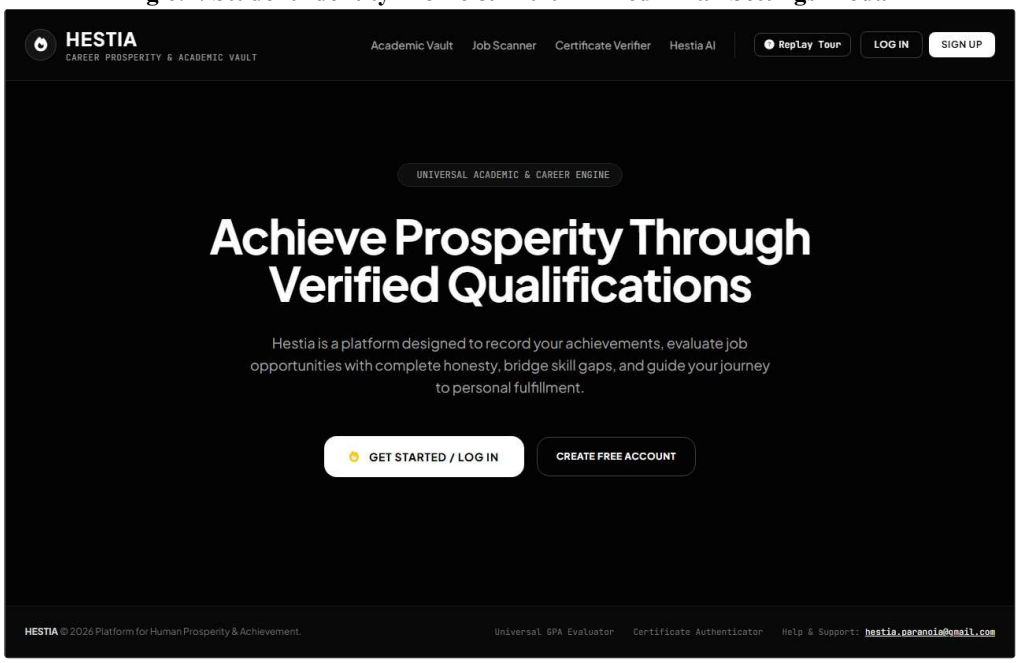


Fig 6.3: Universal Academic Landing Interface & Feature Navigation

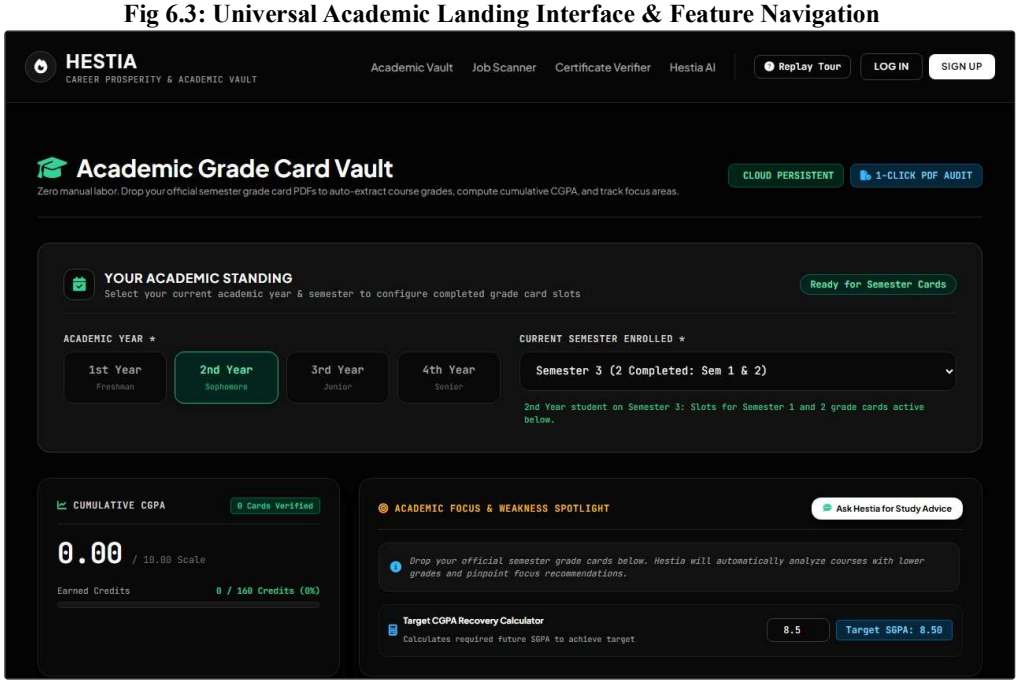


Fig 6.4: Academic Grade Card Vault & Target CGPA Calculator

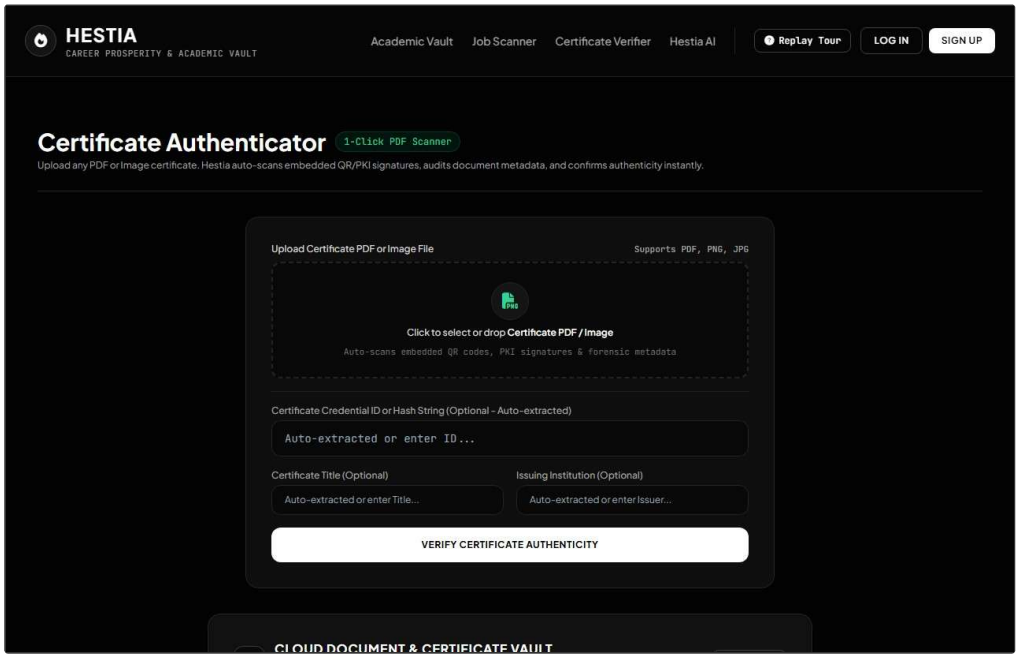


Fig 6.5: 1-Click PDF Certificate Authenticator & QR Validation Interface

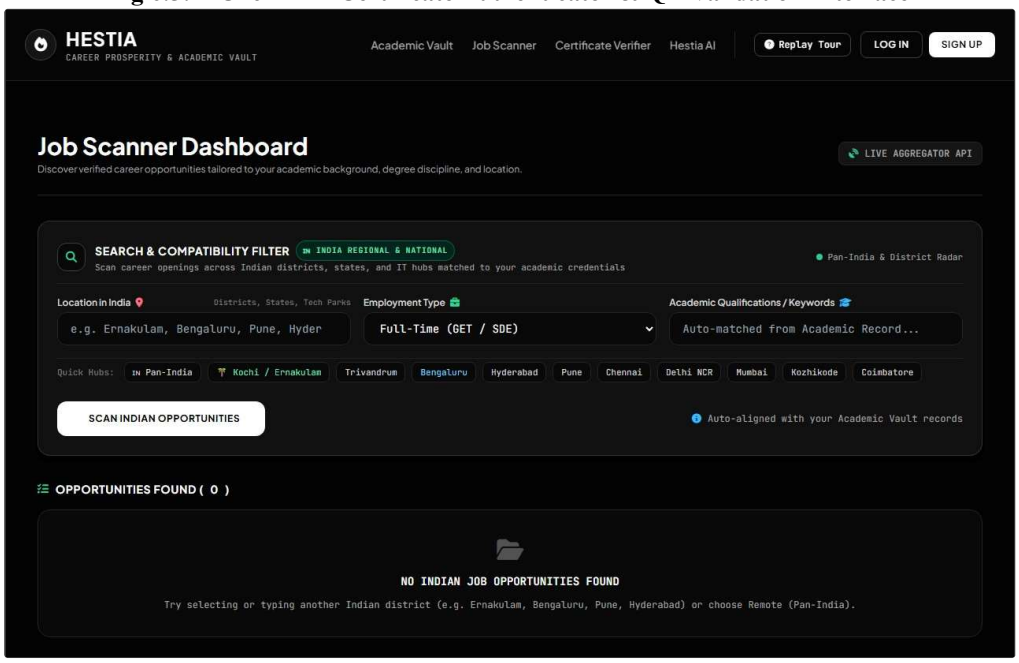


Fig 6.6: Pan-India District Job Radar & LPA Salary Matching

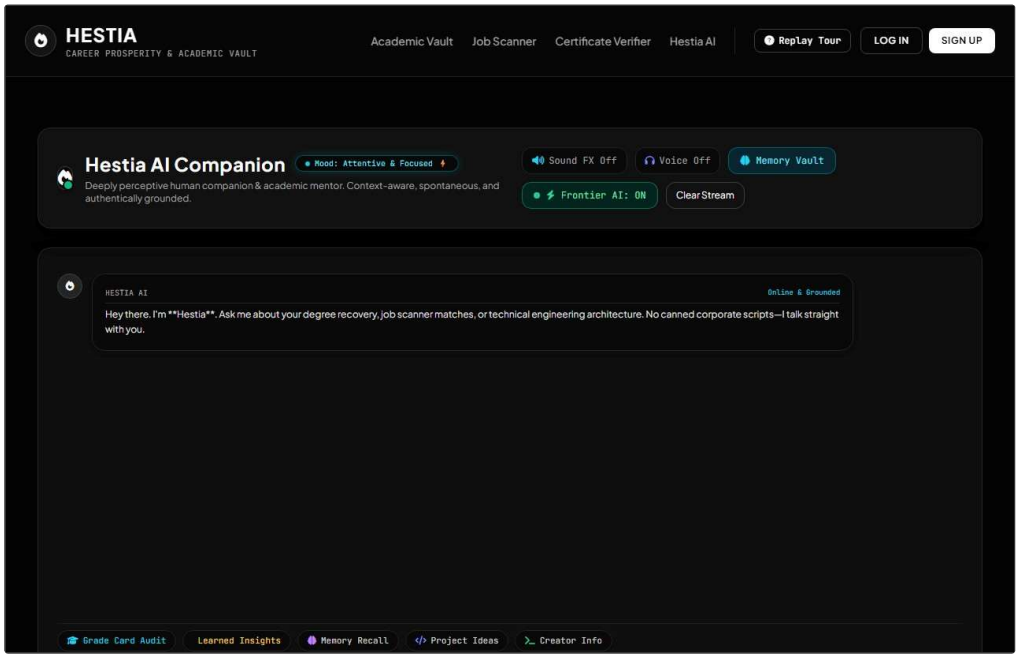


Fig 6.7: Hestia Frontier AI Career Companion & Neural Chatbot

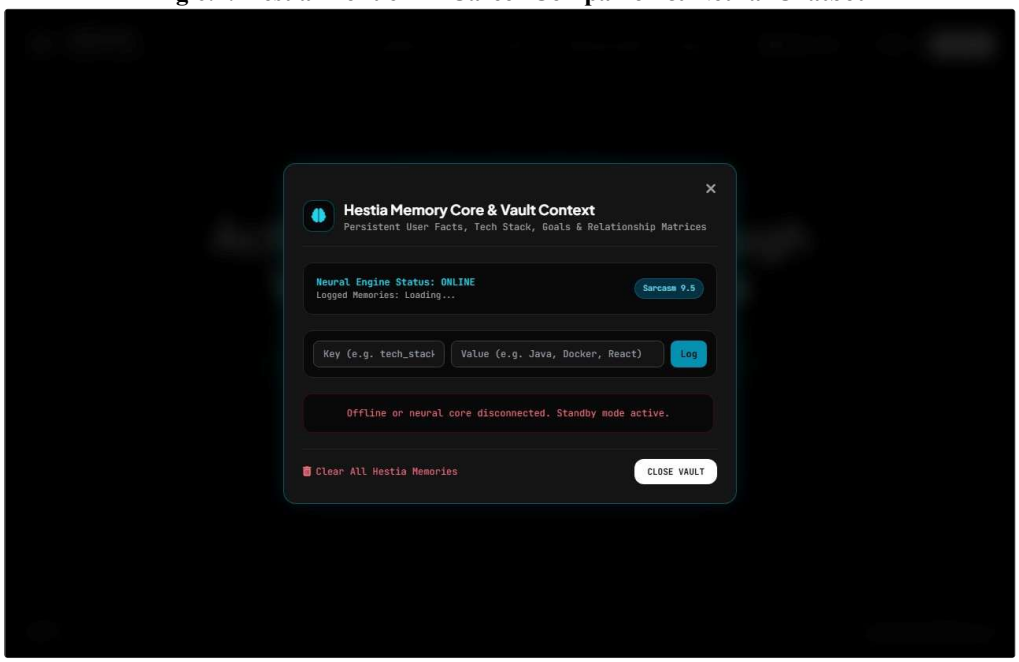


Fig 6.8: Hestia Memory Core Modal & Extracted Fact Vault